

Premier University, Chittagong.

Department of Computer Science and Engineering.



Final Project Report

on

Instacart Online Grocery Basket Analysis

Course: Big Data Analytics Laboratory.

Course Code: CSE 4346.

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1. Introduction

The rapid growth of digital applications has resulted in the generation of enormous amounts of data every day. Online shopping platforms, social media services, financial systems, healthcare organizations, and streaming platforms continuously produce data that requires efficient storage and processing techniques. Conventional database systems often become inefficient when handling datasets containing millions of records, creating the need for distributed computing technologies.

Hadoop has become one of the most popular frameworks for Big Data processing because it combines distributed storage and distributed computation into a single ecosystem. HDFS divides large files into multiple blocks and stores them across different nodes, while MapReduce processes those blocks in parallel to improve performance. Apache Pig extends Hadoop by providing a simpler scripting approach for performing large-scale data analysis without writing complex Java programs for every operation.

This project uses the Instacart Online Grocery Basket Analysis Dataset, which contains real grocery shopping records from Instacart customers. The dataset includes information about customer orders, purchased products, departments, and aisles, making it suitable for analyzing customer purchasing behavior. By implementing both Java MapReduce programs and Apache Pig scripts, the project demonstrates different approaches to solving analytical problems on large datasets. The completed experiments illustrate how distributed computing can efficiently process hundreds of millions of records while producing meaningful business insights.

2. Project Objectives

The primary objectives of this project are:

- Store a large dataset using Hadoop Distributed File System (HDFS).
- Process customer purchasing records using Java MapReduce.
- Perform high-level data analysis using Apache Pig.
- Generate distributed outputs for different analytical tasks.
- Compare the use of Java MapReduce and Pig Latin for Big Data processing.
- Demonstrate practical implementation of Hadoop ecosystem components.

3. Dataset Description

The project uses the Instacart Online Grocery Basket Analysis Dataset collected from Kaggle. The complete dataset contains over 700 MB of grocery shopping records and includes millions of customer purchase entries. Although the original dataset contains six CSV files, three primary files were selected for this project because they provide sufficient information for implementing MapReduce and Pig operations.

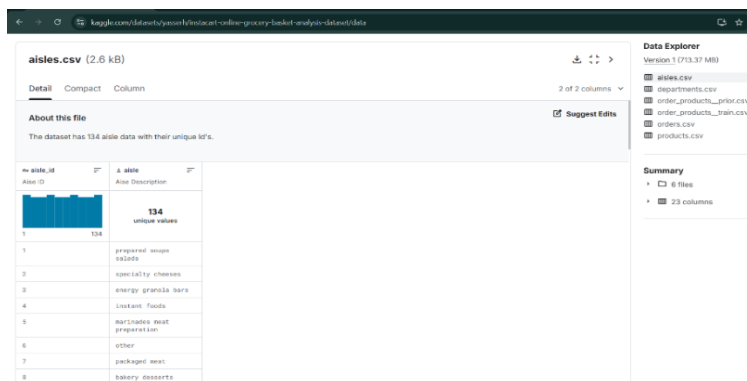
File	Purpose
orders.csv	Customer order information
order_products__prior.csv	Previously purchased products
products.csv	Product information

Selected Dataset Summary

Attribute	Value
Source	Kaggle
Total Dataset Size	713 MB
Original Files	6 CSV Files
Files Used	3
Data Type	Customer Grocery Orders

The selected files contain customer order history, purchased product records, and product metadata, allowing different analytical tasks to be implemented.

Figure 1: Instacart Dataset Source



4. Hadoop Environment Setup

Before executing the project, the Hadoop environment was configured on the local machine. The required software components included Java Development Kit (JDK), Apache Hadoop, Hadoop Distributed File System (HDFS), YARN, and Apache Pig.

After configuring the environment variables and starting Hadoop services, the running processes were verified using the `jps` command. The successful execution confirmed that the NameNode and DataNode services were running correctly, allowing HDFS operations to be performed throughout the project.

Required Components

- Java JDK
- Apache Hadoop
- HDFS
- YARN
- Apache Pig

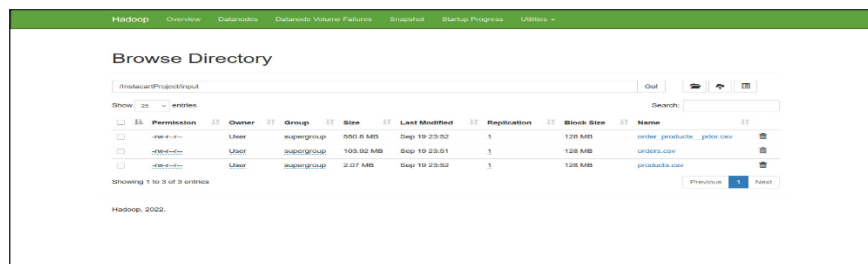
Figure 2: Hadoop Services Verification

```
C:\hadoop\sbin>jps
928 NameNode
22856 DataNode
3736 Jps

C:\hadoop\sbin>_
```

5. Dataset Upload to HDFS

The selected dataset files were uploaded into the HDFS directory named `InstacartProject`. Creating a separate project directory inside HDFS helps organize both the input datasets and generated outputs. The uploaded files were verified using the HDFS listing command to ensure that the required files were successfully stored before executing any analytical tasks.



File	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
☐	drwxr-xr-x	User	supergroup	560.0 MB	Sep 10 23:52	1	128 MB	instacart_project_data.csv
☐	drwxr-xr-x	User	supergroup	100.02 MB	Sep 10 23:51	1	128 MB	instacart_project_orders.csv
☐	drwxr-xr-x	User	supergroup	2.07 MB	Sep 10 23:52	1	128 MB	instacart_project_products.csv

Figure 3: HDFS Project Directory

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject
Found 2 items
drwxr-xr-x - User supergroup      0 2026-09-19 23:52 /InstacartProject/input
drwxr-xr-x - User supergroup      0 2026-09-26 18:13 /InstacartProject/output
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>
```

6. MapReduce Implementation

Three Java MapReduce programs were developed to perform distributed analysis on the Instacart dataset. Each program processes data stored inside HDFS and generates separate output directories.

6.1 MapReduce 1: Orders by Day

Objective

Count the number of customer orders placed on each day of the week.

Description

This program reads orders.csv, extracts the order_dow field, groups records by day, and calculates the total number of orders for each day. The Mapper emits (day,1) pairs while the Reducer aggregates the counts.

MR1 – Mapper Logic

The mapper reads each record from orders.csv, extracts the order_dow field, and emits (day,1). Example: day 0 → (0,1), day 6 → (6,1).

MR1 – Reducer Logic

The reducer groups all values belonging to the same day and sums them to calculate the total number of orders for each day of the week.

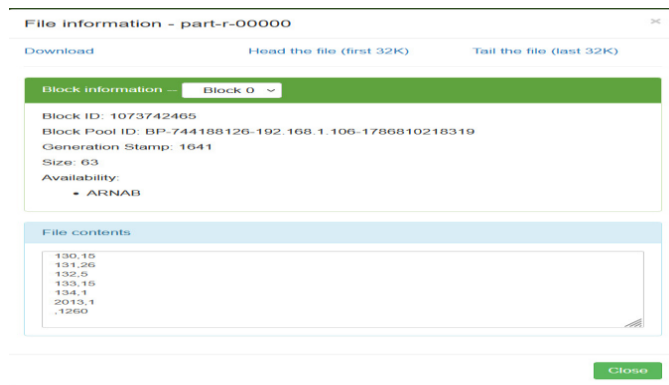
Figure 4: MR1 Output Directory

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject/output/MR1_OrdersByDay
Found 2 items
-rw-r--r-- 1 User supergroup      0 2026-09-19 23:58 /InstacartProject/output/MR1_OrdersByDay/_SUCCESS
-rw-r--r-- 1 User supergroup    63 2026-09-19 23:58 /InstacartProject/output/MR1_OrdersByDay/part-r-00000
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>
```

Figure 5: MR1 Result

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -cat /InstacartProject/output/MR1_OrdersByDay/part-r-00000
0      600905
1      587478
2      467260
3      436972
4      426339
5      453368
6      448761
```

The same output was also verified through the HDFS browser:



Analysis

The generated output shows that customer ordering behavior varies across different days. Some days receive significantly higher order volumes, indicating that shopping activity is not evenly distributed throughout the week.

6.2 MapReduce 2: Orders by Hour

Objective

Analyze customer ordering frequency throughout the day.

Description

The program extracts order_hour_of_day from orders.csv and counts how many orders were placed during each hour.

MR2 – Mapper Logic

The mapper extracts order_hour_of_day from each order and emits (hour,1). Example: hour 10 → (10,1).

MR2 – Reducer Logic

The reducer aggregates all values for each hour and produces the total number of orders placed during every hour of the day.

Figure 6: MR2 Output Directory

```
C:\n26r2\n26r\p62kfo\Iu2f9c9rf-Bj8D9f9-Åu9JYfIc2\Onfbnfz>
-LM-L--L-- J n26r znb6rBtonb 5T2 S0S0-00-50 00:JT \Iu2f9c9rfBto]ecf\onfbnf\WKS~0Lq6r2B\H0NL\b9rf-L-00000
-LM-L--L-- J n26r znb6rBtonb 0 S0S0-00-50 00:JT \Iu2f9c9rfBto]ecf\onfbnf\WKS~0Lq6r2B\H0NL~2NCC22
f0nuq 5 Jf6wz
C:\n26r2\n26r\p62kfo\Iu2f9c9rf-Bj8D9f9-Åu9JYfIc2\Onfbnfz>µqfz qfz -J2 \Iu2f9c9rfBto]ecf\onfbnf\WKS~0Lq6r2B\H0NL
```

Figure 7: MR2 Result

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -cat /InstacartProject/output/MR2_OrdersByHour/part-r-000000
0      22758
1      12398
2      7539
3      5474
4      5527
5      9569
6      30529
7      91868
8      178201
9      257812
10     288418
11     284728
12     272841
13     277999
14     283842
15     283639
16     272553
17     228795
18     182912
19     140569
20     104292
21     78109
22     61468
23     40043
```

The same output was also verified through the HDFS browser:



Analysis

The results indicate that customer activity increases during daytime hours and gradually decreases during late-night hours, reflecting typical grocery shopping behavior.

6.3 MapReduce 3: Top Purchased Products

Objective

Identify the most frequently purchased products.

Description

This program processes `order_products__prior.csv`, groups records by `product_id`, and calculates purchase frequency for each product.

MR3 – Mapper Logic

The mapper reads `order_products__prior.csv`, extracts `product_id`, and emits `(product_id,1)`. Example: product 24852 → `(24852,1)`.

MR3 – Reducer Logic

The reducer sums all purchase counts for each product and generates the purchase frequency used to rank the most frequently purchased products.

Figure 8: MR3 Output Directory


```

products = LOAD '/InstacartProject/input/order_products__prior.csv'
USING PigStorage(',')
AS (
  order_id:int,
  product_id:int,
  add_to_cart_order:int,
  reordered:int
);

reordered_products = FILTER products BY reordered == 1;

STORE reordered_products
INTO '/InstacartProject/output/Pig1_ReorderedProducts'
USING PigStorage(',')|

```

Figure 10: Pig1 Output

```

C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject/output/Pig1_ReorderedProducts
Found 6 items
-rw-r--r--  1 User supergroup          0 2026-09-20 00:20 /InstacartProject/output/Pig1_ReorderedProducts/_SUCCESS
-rw-r--r--  1 User supergroup 78891383 2026-09-20 00:20 /InstacartProject/output/Pig1_ReorderedProducts/part-m-00000
-rw-r--r--  1 User supergroup 79130251 2026-09-20 00:20 /InstacartProject/output/Pig1_ReorderedProducts/part-m-00001
-rw-r--r--  1 User supergroup 78912960 2026-09-20 00:20 /InstacartProject/output/Pig1_ReorderedProducts/part-m-00002
-rw-r--r--  1 User supergroup 78956159 2026-09-20 00:20 /InstacartProject/output/Pig1_ReorderedProducts/part-m-00003

```

Output on Hdfs:

File information - part-m-00000

Download Head the file (first 32K) Tail the file (last 32K)

Block information - Block 0

Block ID: 1073742500
 Block Pool ID: BP-744188126-192.168.1.106-1786810218319
 Generation Stamp: 1676
 Size: 78891383
 Availability:
 • ARNAB

File contents

0	6009005
1	587478
2	467260
3	436972
4	426339
5	453368
6	448761

Close

Analysis

The output represents products that customers purchased repeatedly, providing insight into customer loyalty toward specific grocery items.

7.2 Pig 2: First Cart Products

Objective

Extract products added first to shopping carts.

Description

The script filters records where `add_to_cart_order = 1`, representing the first product added during each shopping session.

Pig2 – Pig Script Logic

1. Load `order_products__prior.csv`.
2. Filter records where `add_to_cart_order = 1`.
3. Store the first-cart products into HDFS.
4. Verify the generated output.

Figure 11: Pig2 Script

```
products = LOAD '/InstacartProject/input/order_products__prior.csv'
USING PigStorage(',')
AS (
  order_id:int,
  product_id:int,
  add_to_cart_order:int,
  reordered:int
);

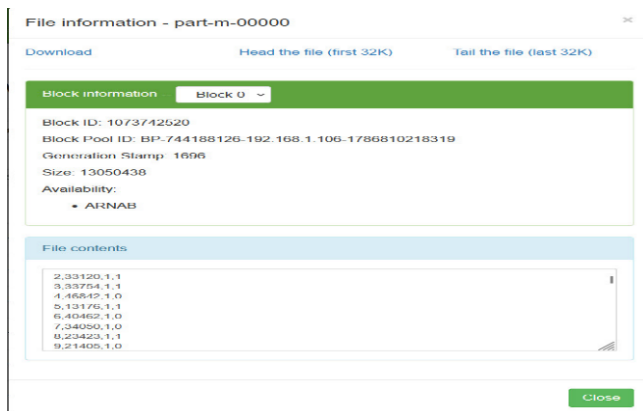
first_products = FILTER products BY add_to_cart_order == 1;

STORE first_products
INTO '/InstacartProject/output/Pig2_FirstCartProducts'
USING PigStorage(',');
```

Figure 12: Pig2 Output

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject/output/Pig2_FirstCartProducts
Found 6 items
-rw-r--r--  1 User supergroup          0 2026-09-20 00:30 /InstacartProject/output/Pig2_FirstCartProducts/_SUCCESS
-rw-r--r--  1 User supergroup 13050438 2026-09-20 00:30 /InstacartProject/output/Pig2_FirstCartProducts/part-m-00000
-rw-r--r--  1 User supergroup 13056959 2026-09-20 00:30 /InstacartProject/output/Pig2_FirstCartProducts/part-m-00001
-rw-r--r--  1 User supergroup 13054546 2026-09-20 00:30 /InstacartProject/output/Pig2_FirstCartProducts/part-m-00002
-rw-r--r--  1 User supergroup 13068352 2026-09-20 00:30 /InstacartProject/output/Pig2_FirstCartProducts/part-m-00003
-rw-r--r--  1 User supergroup 3958233 2026-09-20 00:30 /InstacartProject/output/Pig2_FirstCartProducts/part-m-00004
```

Output on Hdfs:



Analysis

The results highlight products that customers tend to prioritize when beginning their shopping sessions.

7.3 Fig 3: Top Purchased Products

Objective

Rank products by purchase frequency.

Description

The script groups records by product_id, counts purchases, and sorts the results in descending order.

Pig3 – Pig Script Logic

1. Load the dataset.
2. Group records by product_id.
3. Count purchases for every product.
4. Sort the results in descending order.
5. Store the ranked output into HDFS.

Figure 13: Pig3 Script

```
products = LOAD '/InstacartProject/input/order_products_prior.csv'
USING PigStorage(',')
AS (
  order_id:int,
  product_id:int,
  add_to_cart_order:int,
  reordered:int
);

grp = GROUP products BY product_id;

cnt = FOREACH grp GENERATE group AS product_id, COUNT(products) AS total_purchases;

ordered = ORDER cnt BY total_purchases DESC;

STORE ordered
INTO '/InstacartProject/output/Pig3_TopPurchasedProducts'
USING PigStorage(',');
```

Figure 14: Pig3 Output

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject/output/Pig3_TopPurchasedProducts
Found 2 items
-rw-r--r-- 1 User supergroup 0 2026-09-20 00:44 /InstacartProject/output/Pig3_TopPurchasedProducts/_SUCCESS
-rw-r--r-- 1 User supergroup 454460 2026-09-20 00:44 /InstacartProject/output/Pig3_TopPurchasedProducts/part-r-00000
```

Output on Hdfs:



Analysis

The script performs aggregation and sorting using fewer lines of code than Java MapReduce while producing similar analytical results.

7.4 Pig 4: Weekend Orders

Objective

Extract orders placed on weekends.

Description

The script filters orders.csv where order_dow represents Saturday or Sunday.

Pig4 – Pig Script Logic

- 1. Load orders.csv.
- 2. Filter records where order_dow = 0 or order_dow = 6.
- 3. Store the weekend orders into HDFS.
- 4. Verify the output directory.

Figure 15: Pig4 Script

```

orders = LOAD '/InstacartProject/input/orders.csv'
USING PigStorage(',')
AS (
  order_id:int,
  user_id:int,
  eval_set:chararray,
  order_number:int,
  order_dow:int,
  order_hour_of_day:int,
  days_since_prior_order:chararray
);

weekend = FILTER orders BY (order_dow == 0) OR (order_dow == 6);

STORE weekend
INTO '/InstacartProject/output/Pig4_WeekendOrders'
USING PigStorage(',');

```

Figure 16: Pig4 Output

```

C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject/output/Pig4_WeekendOrders
Found 2 items
-rw-r--r--  1 User supergroup          0 2026-09-26 17:46 /InstacartProject/output/Pig4_WeekendOrders/_SUCCESS
-rw-r--r--  1 User supergroup 33241282 2026-09-26 17:46 /InstacartProject/output/Pig4_WeekendOrders/part-m-000000

```

Output On Hdfs:

File information - part-m-000000

Download Head the file (first 32K) Tail the file (last 32K)

Block information -- Block 0

Block ID: 1073742576
 Block Pool ID: BP-744188126-192.168.1.106-1786810218319
 Generation Stamp: 1752
 Size: 33241282
 Availability:
 • ARNAB

File contents

```

2710558,3,prior,5,0,17,12,0
1972919,3,prior,6,0,16,7,0
1839752,3,prior,7,0,15,7,0
3225766,3,prior,8,0,17,7,0
3160850,3,prior,9,0,16,7,0
521107,3,prior,11,0,16,11,0
3343014,4,prior,1,6,11
1909121,5,prior,2,0,16,11,0

```

Close

Analysis

The resulting dataset contains only weekend shopping activity, making it useful for analyzing customer behavior during weekends separately.

7.5 Pig 5: Department-wise Product Count

Objective

Count the number of products available in each department.

Description

The script loads products.csv, removes the header row, groups products by department, and calculates product counts.

Pig5 – Pig Script Logic

1. Load products.csv.
2. Group products by department_id.
3. Count the number of products in each department.
4. Store the department-wise counts into HDFS.
5. Verify the generated output.

Figure 17: Pig5 Script

```
raw = LOAD '/InstacartProject/input/products.csv'
USING PigStorage(',')
AS (
  product_id:chararray,
  product_name:chararray,
  aisle_id:chararray,
  department_id:chararray
);

-- Header remove
products = FILTER raw BY product_id != 'product_id';

-- Convert data types
clean = FOREACH products GENERATE
  (int)product_id AS product_id,
  product_name AS product_name,
  (int)aisle_id AS aisle_id,
  (int)department_id AS department_id;

grp = GROUP clean BY department_id;

result = FOREACH grp GENERATE
  group AS department_id,
  COUNT(clean) AS total_products;

STORE result
INTO '/InstacartProject/output/Pig5_DepartmentCount'
USING PigStorage(',');
```

Figure 18: Pig5 Output

```
C:\Users\User\Desktop\Instacart-BigData-Analytics\Outputs>hdfs dfs -ls /InstacartProject/output/Pig5_DepartmentCount
Found 2 items
-rw-r--r--  1 User supergroup          0 2026-09-26 18:13 /InstacartProject/output/Pig5_DepartmentCount/_SUCCESS
-rw-r--r--  1 User supergroup      809 2026-09-26 18:13 /InstacartProject/output/Pig5_DepartmentCount/part-r-00000
```

Output On Hdfs:

File information - part-r-00000

[Download](#) [Head the file \(first 32K\)](#) [Tail the file \(last 32K\)](#)

Block information

Block 0

Block ID: 1073742633
Block Pool ID: BP-744188126-192.168.1.106-1786810218319
Generation Stamp: 1809
Size: 809
Availability:
• ARNAB

File contents

2,33120,1,1
3,33754,1,1
4,46842,1,0
5,13176,1,1
6,40462,1,0
7,34050,1,0
8,23423,1,1
9,21405,1,0

Close

Analysis

The output summarizes how products are distributed across different departments, providing an overview of inventory composition.

8. Results and Discussion

The project successfully demonstrates the practical implementation of Hadoop's distributed computing environment using the Instacart grocery dataset. Both Java MapReduce and Apache Pig completed their assigned analytical tasks by processing data directly from HDFS and storing the generated outputs into separate project directories.

The three MapReduce programs produced different business-oriented insights. MR1 calculated the number of customer orders for each day of the week, showing that shopping activity changes across different days. MR2 analyzed order frequency by hour and revealed that customer activity is concentrated during daytime hours rather than late-night periods. MR3 identified the most frequently purchased products by counting purchase occurrences from previous order records, helping highlight popular grocery items.

The Pig implementations further simplified large-scale data processing through concise Pig Latin scripts. Pig1 extracted reordered products, making it possible to observe repeat purchasing behavior. Pig2 identified products that customers typically added first to their shopping carts, while Pig3 ranked products according to total purchase frequency. Pig4 successfully filtered weekend orders, creating a separate dataset for Saturday and Sunday shopping behavior. Finally, Pig5 calculated department-wise product counts, providing a summary of product distribution across different departments.

Overall, the project demonstrates that Java MapReduce is suitable for customized distributed computations, whereas Apache Pig performs filtering, grouping, counting, and sorting operations with much simpler code. The generated outputs confirm that Hadoop can efficiently process large datasets and produce meaningful analytical results through distributed execution.

Summary of Generated Outputs

Program Result

MR1	Orders grouped by day
MR2	Orders grouped by hour
MR3	Top purchased products
Pig1	Reordered products
Pig2	First cart products

Pig3	Top purchased products
Pig4	Weekend orders
Pig5	Department-wise product count

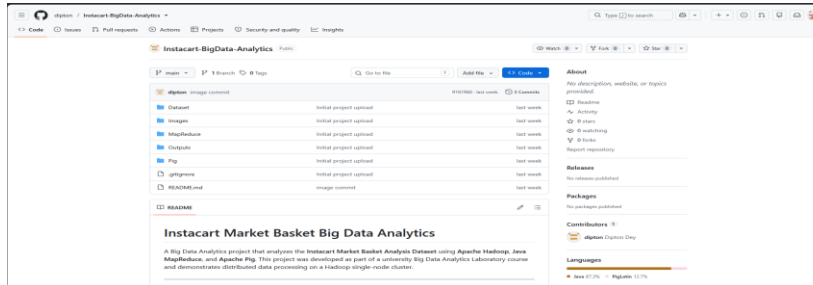
9. GitHub Repository

The complete project, including the Java source code, Pig scripts, project structure, documentation, and generated outputs, was uploaded to GitHub for version control and future reference.

Repository:

https://github.com/dipton/Instacart-BigData-Analytics_.git

Figure 19: GitHub Repository



10. Conclusion

This project successfully demonstrates how Hadoop ecosystem technologies can be used to process and analyze large-scale datasets through distributed computing. By using the Instacart Online Grocery Basket dataset, practical experience was gained in storing data inside HDFS, executing Java MapReduce programs, and performing high-level analysis using Apache Pig. The implemented experiments covered several real-world analytical scenarios, including customer ordering behavior, product popularity, reordered products, weekend shopping activity, and department-wise product distribution.

The comparison between Java MapReduce and Apache Pig also highlights their complementary roles within the Hadoop ecosystem. Java MapReduce provides greater flexibility for building customized distributed algorithms, while Pig Latin significantly reduces development time for common analytical operations such as filtering, grouping, counting, and sorting.

Overall, this project provides valuable hands-on experience with Big Data technologies and demonstrates how Hadoop can efficiently process millions of records that would be difficult to manage using traditional data processing methods. The knowledge gained from this implementation forms a strong foundation for working with large-scale data analytics, distributed systems, and future Big Data applications.

References

1. Kaggle. Instacart Online Grocery Basket Analysis Dataset.
2. Apache Hadoop Official Documentation.
3. Apache Pig Official Documentation.
4. Oracle Java Documentation.
5. Hadoop Distributed File System (HDFS) User Guide.